Representations: Latent, Implicit, Explicit

AI 4 Life Science Summer School - Aug 19

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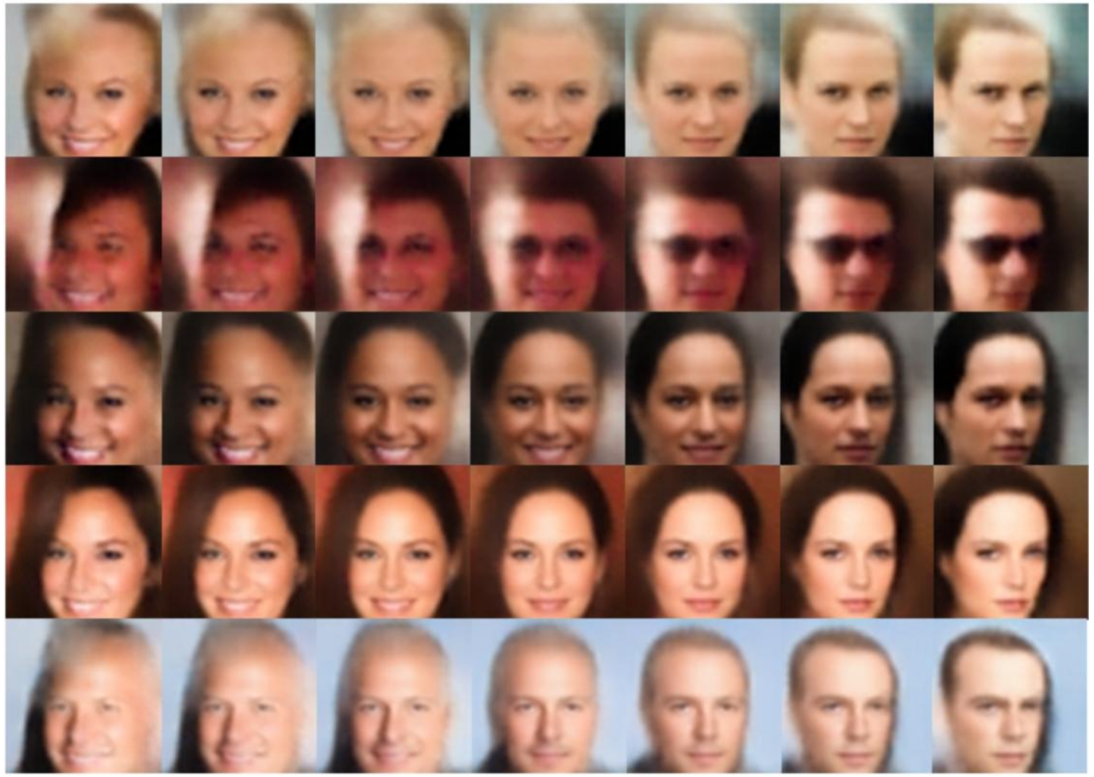
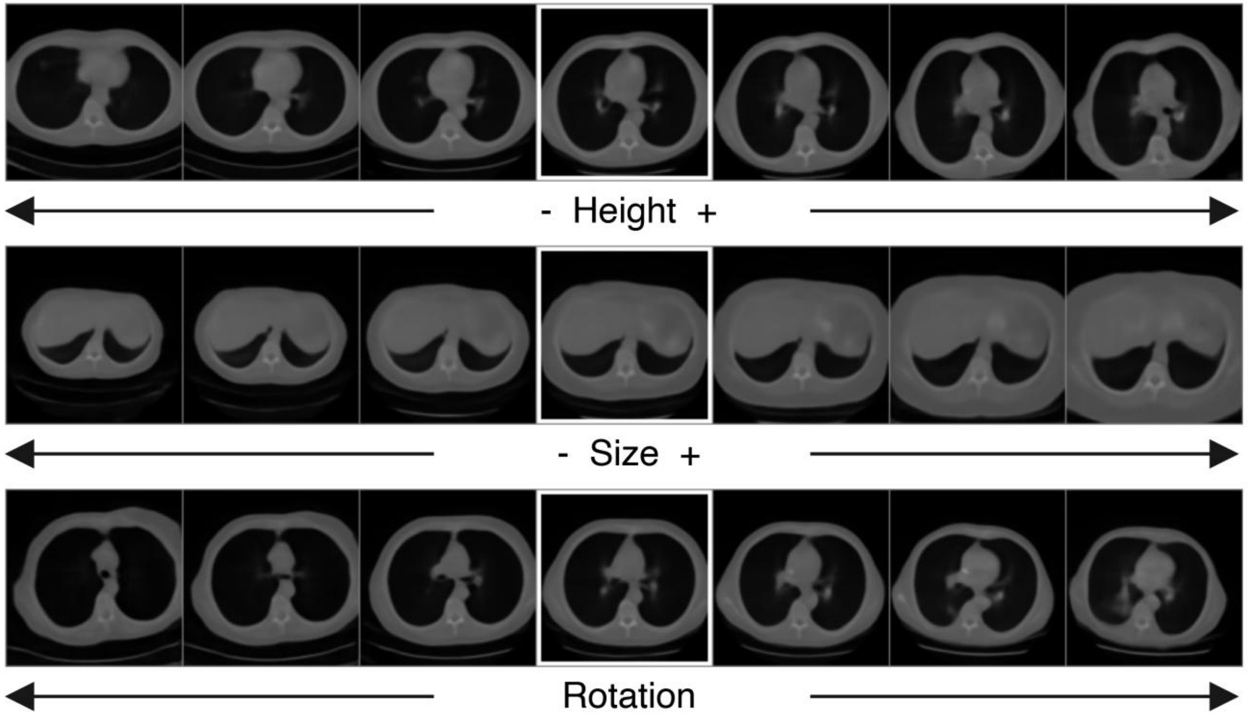
What we have? Vs. What we want?

- The data itself (images)
- Domain knowledge of:
  - The sampled distribution
  - The subject
- Assumptions

- Approximate features of supervised model
- General features for any task
- Manifold where data lies
- Cluster/biomarker discovery

Latent Representations





I. Higgins, et al., ICLR. 2017.

J. Schön, et al., DGM4MICCAI. 2022.

Medical Classification

Table 2 2D classification linear probing results comparing baseline and DINOv3 series on the NIH-14 and RSNA-Pneumonia datasets. All models use an input resolution of 256x256. For each metric, the highest performing method is marked in **bold**, and the second highest is <u>underlined</u>.

Methods	NIH-14				RSNA-Pneumonia			
	AUC	ACC	Precision	Recall	AUC	ACC	Precision	Recall
BiomedCLIP [17]	0.7771	0.4820	**0.5454**	0.5643	**0.8831**	**0.8374**	**0.6368**	<u>0.8026</u>
DINOv3-S [11]	0.7788	**0.4838**	0.5419	**0.5791**	0.8667	0.8221	0.6048	**0.8156**
DINOv3-B [11]	<u>0.7833</u>	<u>0.4823</u>	<u>0.5446</u>	0.5753	0.8666	<u>0.8274</u>	<u>0.6227</u>	0.7679
DINOv3-L [11]	**0.7865**	0.4674	0.5355	<u>0.5779</u>	<u>0.8708</u>	0.8209	0.5972	0.7744

C. Liu, et al., arXiv:2509.06467. 2026.

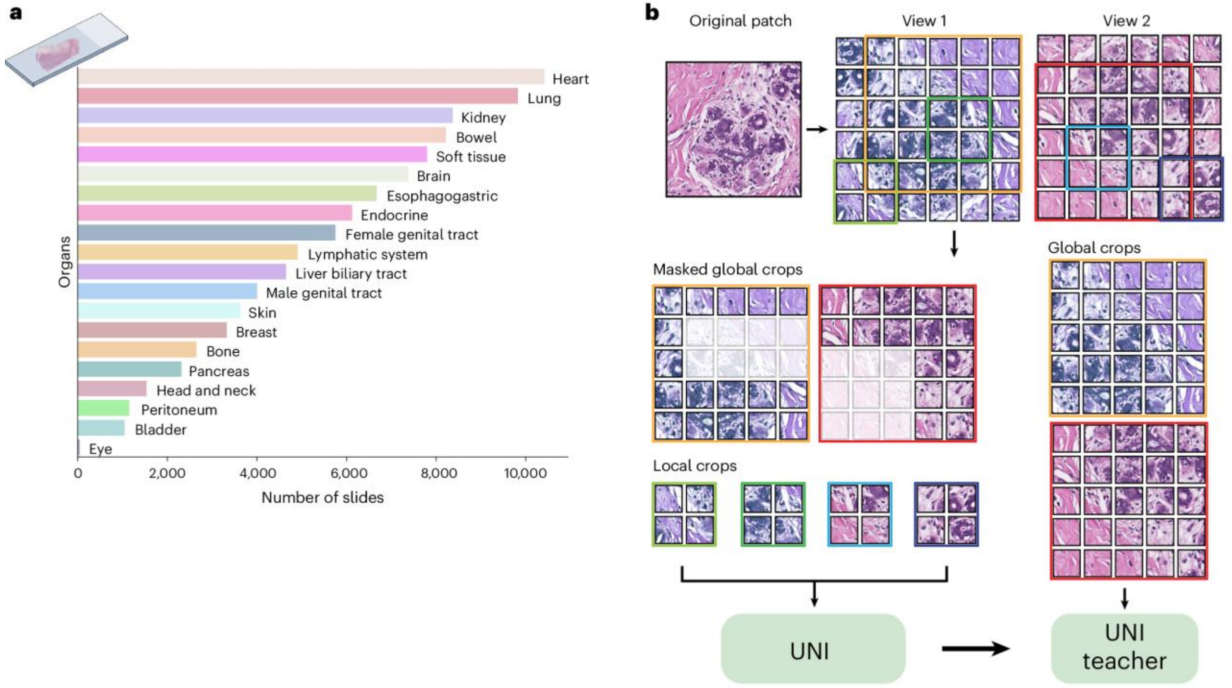
Medical Classification

Table 3 In-domain tumour detection on Camelyon16. Patch features are aggregated with ABMIL. Models are trained on the Camelyon16 training set and evaluated on its test set. The highest results are in **bold** and the second highest are <u>underlined</u>.

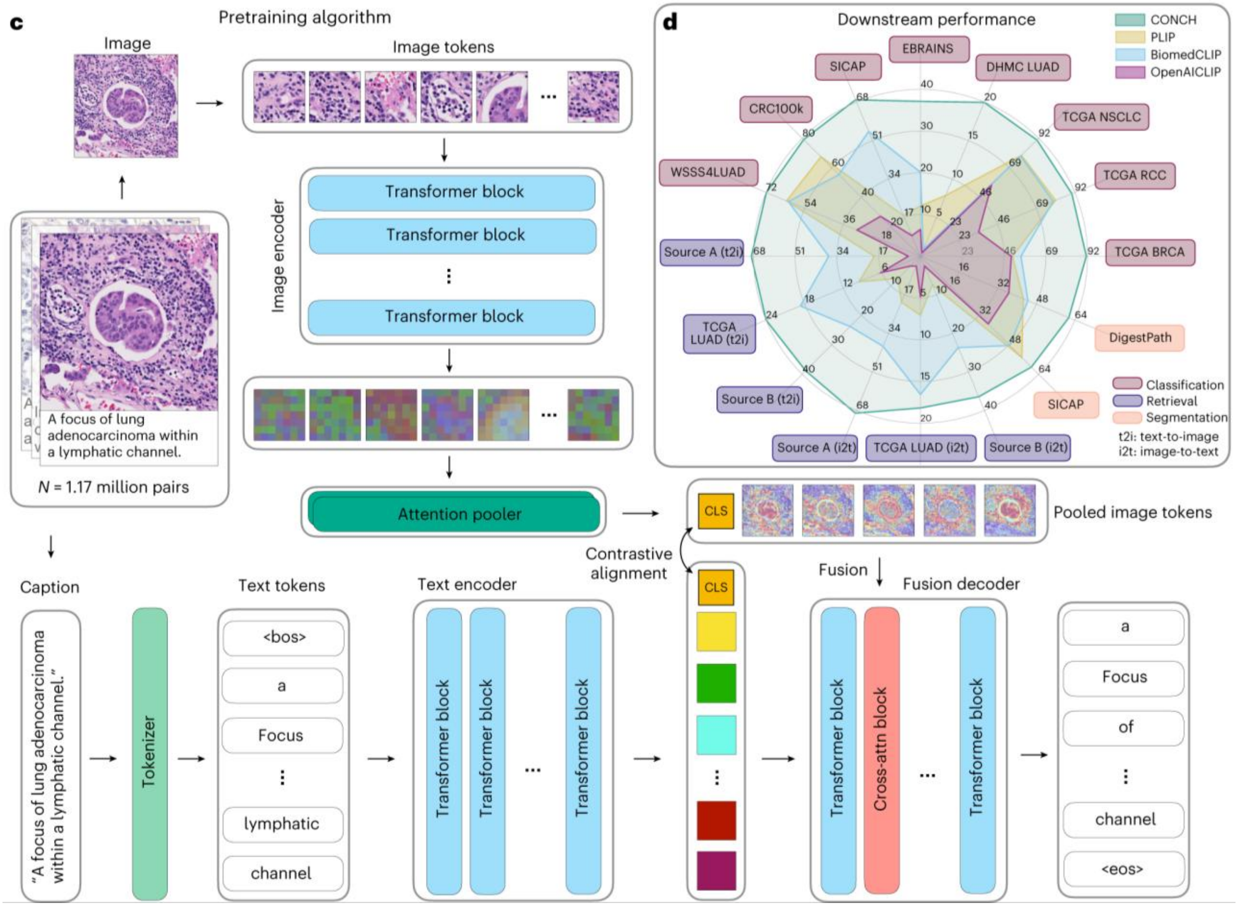
Patch Encoder	**Camelyon16 → Camelyon16**			
	AUC	**ACC**	**Precision**	**Recall**
ResNet50 (ImageNet) [45]	0.842	0.713	0.594	0.776
UNI [44]	**0.965**	**0.951**	**0.959**	**0.938**
CONCH [15]	<u>0.961</u>	<u>0.944</u>	<u>0.956</u>	<u>0.928</u>
DINOv3-S [11]	0.840	0.847	0.898	0.682
DINOv3-B [11]	0.805	0.800	0.834	0.629

C. Liu, et al., arXiv:2509.06467. 2026.

Medical Classification



UNI

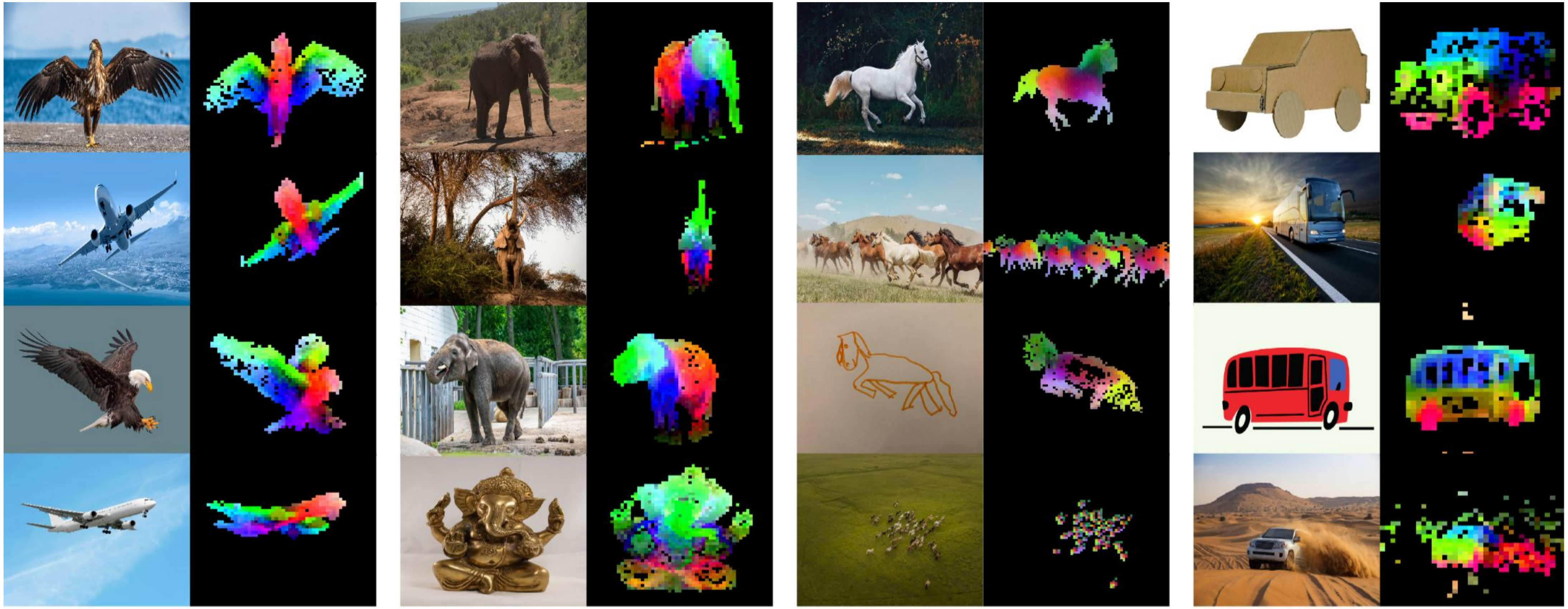


CONCH

R. Chen, et al., Nature Medicine. 2024.

M. Lu, et al., Nature Medicine. 2024.

Contextualized Embeddings



*M. Oquab, et al., TMLR. 2024

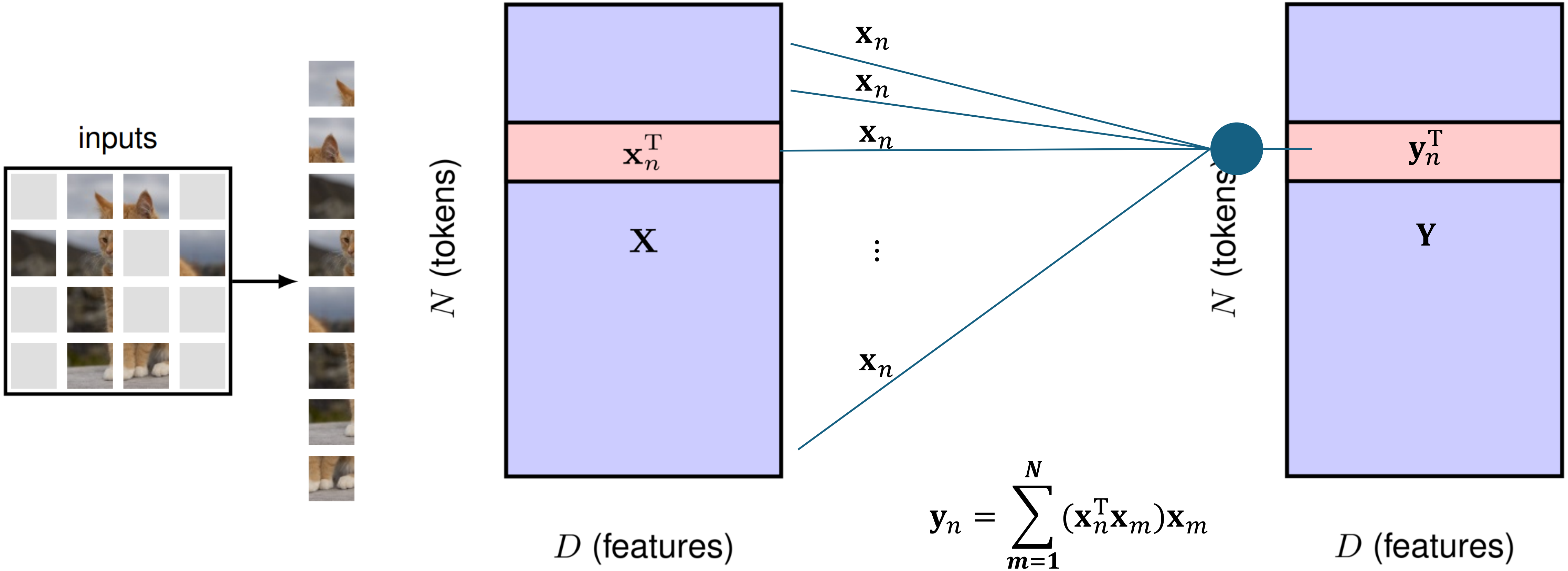
Medical Segmentation Decathlon

Methods	Task01 (Brain)	Task02 (Heart)	Task03 (Liver)	Task04 (Hippo.)	Task05 (Prostate)	Task06 (Lung)	Task07 (Pancreas)	Task08 (Hepatic)	Task09 (Spleen)	Task10 (Colon)	Average
Supervised Learning Methods											
3D U-Net [85]	72.4	81.3	91.2	76.8	82.1	67.9	71.5	55.3	87.6	42.1	72.8
V-Net [86]	71.8	83.7	90.8	78.2	84.3	66.4	73.2	57.1	89.2	41.8	73.7
nnU-Net [84]	**78.9**	**89.4**	**96.2**	**84.1**	**91.3**	**75.8**	**82.7**	**67.9**	**94.8**	**52.6**	**81.4**
TransUNet [71]	74.2	85.1	93.4	79.6	86.7	70.3	76.8	59.4	91.2	45.7	76.2
SwinUNETR [73]	<u>76.5</u>	<u>87.3</u>	<u>94.7</u>	<u>81.2</u>	<u>88.9</u>	<u>72.6</u>	<u>78.9</u>	<u>62.1</u>	<u>92.8</u>	<u>47.3</u>	<u>78.2</u>
UNETR [87]	75.1	86.2	93.9	80.4	87.5	71.8	77.4	60.7	91.9	46.2	77.1
Self-Supervised Methods (Linear Fine-tuning)											
MAE-ViT-B/16 [88]	62.1	73.8	82.3	68.4	75.2	61.4	66.8	48.9	81.2	35.4	65.6
MAE-ViT-L/16 [88]	64.5	76.2	**84.1**	71.2	78.1	64.1	69.3	52.1	84.8	38.7	68.3
SimCLR [89]	58.9	70.1	79.8	64.7	72.5	57.8	63.2	45.1	77.4	31.9	62.1
MoCo-v3 [90]	61.3	73.2	81.6	67.9	74.8	60.9	66.1	48.2	80.6	35.1	64.8
SwAV [91]	60.2	71.8	80.4	66.1	73.6	59.1	64.7	46.8	78.9	33.2	63.5
BYOL [92]	60.8	72.5	80.9	67.3	74.1	60.2	65.4	47.5	79.7	34.6	64.2
DINOv3-S [11]	65.2	77.1	<u>83.8</u>	72.6	78.9	65.8	70.2	53.4	82.9	40.1	69.0
DINOv3-B [11]	**66.8**	**78.2**	**84.1**	**75.3**	<u>79.8</u>	**73.1**	**78.9**	**64.8**	<u>86.4</u>	**49.1**	**71.3**
DINOv3-L [11]	<u>65.9</u>	<u>77.6</u>	83.5	<u>73.8</u>	**80.5**	<u>72.4</u>	<u>78.2</u>	<u>63.7</u>	**91.2**	<u>47.8</u>	<u>71.0</u>

C. Liu, et al., arXiv:2509.06467. 2026.

Masked Autoencoder

*K. He, et al., CVPR. 2022

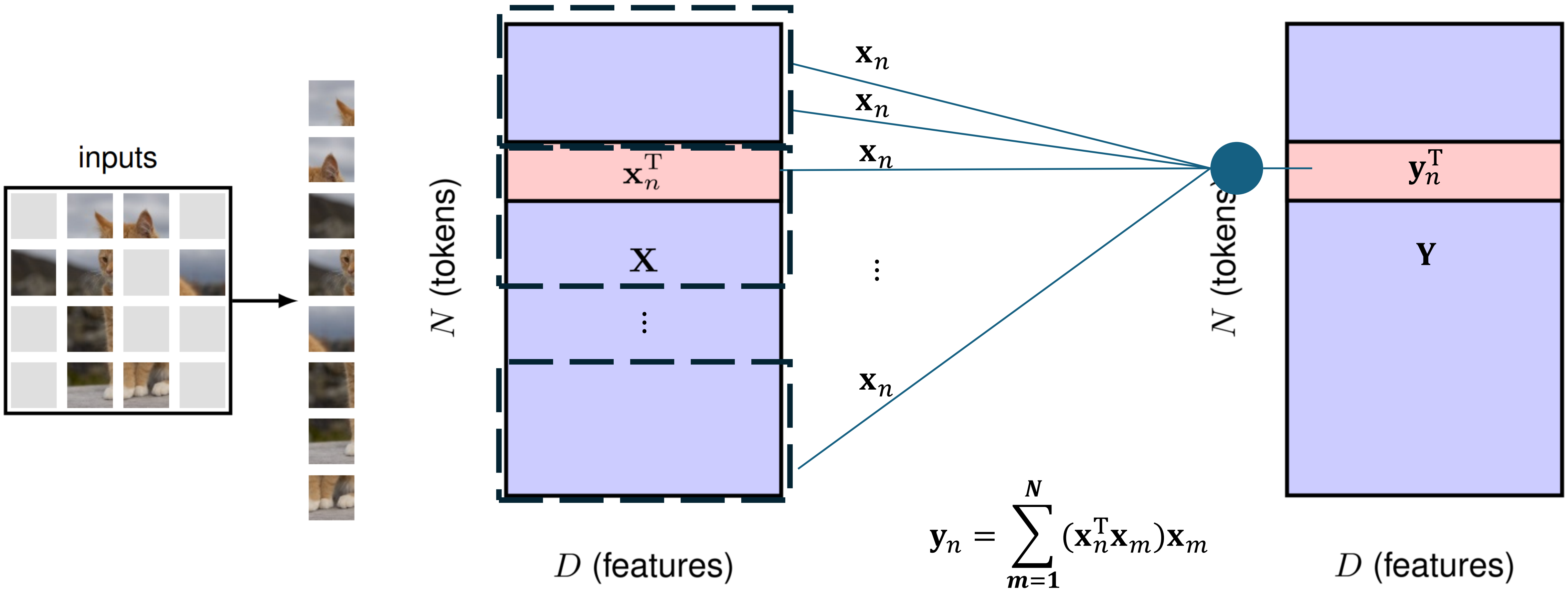


$$\mathbf{y}_n = \sum_{m=1}^{N} (\mathbf{x}_n^{\mathrm{T}} \mathbf{x}_m) \mathbf{x}_m$$

C.M. Bishop and H. Bishop, Deep Learning: Foundations and Concepts. 2023.

Swin Transformer

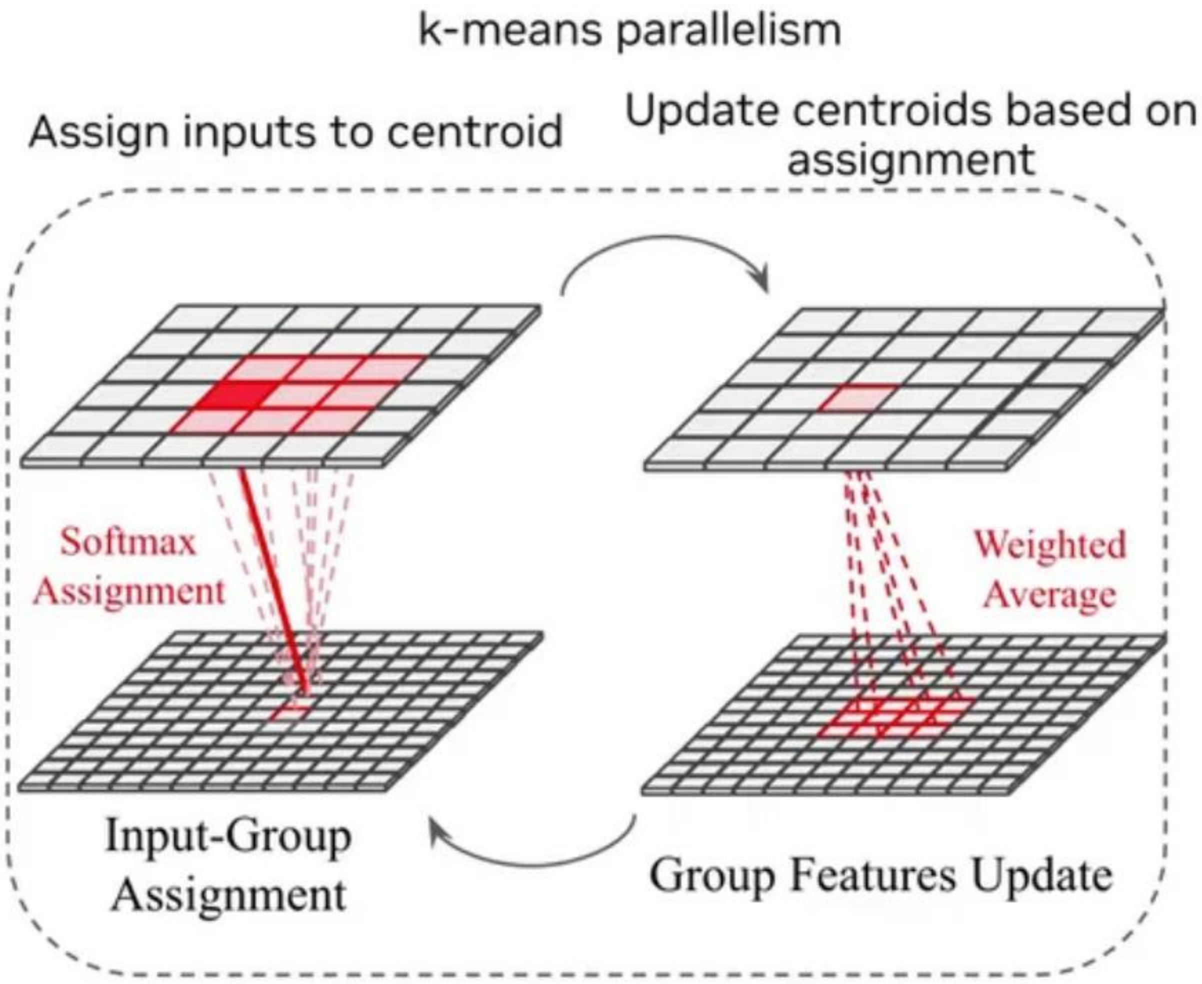
*Z. Liu, et al., ICCV. 2021



C.M. Bishop and H. Bishop, Deep Learning: Foundations and Concepts. 2023.

Native **Se**gmentation Vision **Tra**nsformer (SeNaTra)

*G. Braso, et al., NeurIPS. 2025



Native **Se**gmentation Vision **Tra**nsformer (SeNaTra)

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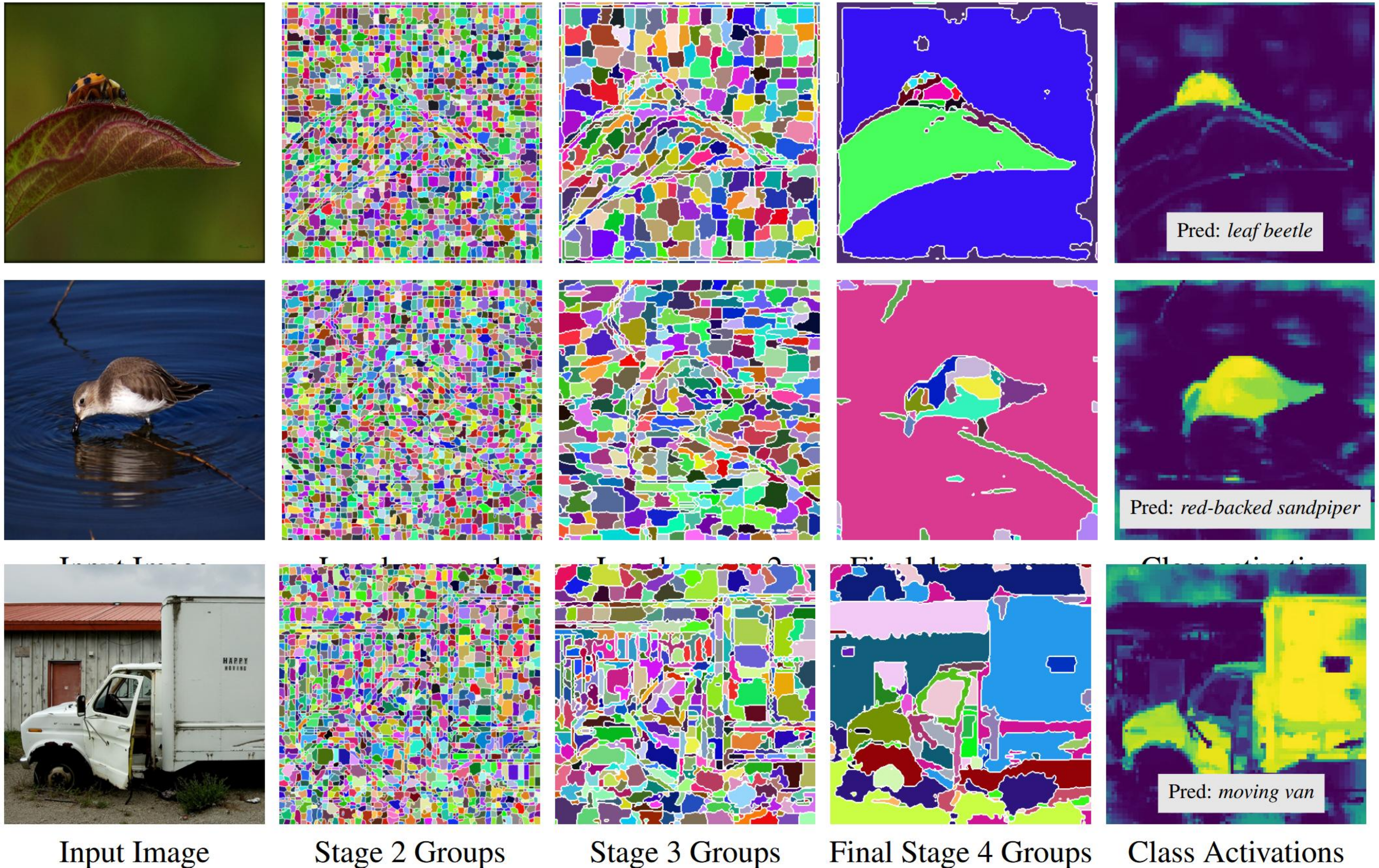

Tutorial #3

AI 4 Life Science - Unsupervised Representation Learning - Tutorial 2: Self-supervision

Setup and Groups

You can choose to remain in the same groups as last time, or repeat the steps from last time with new group members

Form groups of 3, ideally with people you do not work with regularly. Take 5 min each to explain a bit about your work:

1. What is the focus of your work?
2. What is the limitation or most difficult technical aspect?
3. Could unsupervised learning fit into your project in any way? If so, where could it improve things. If not, why is it unhelpful and what do you wish it could do?

https://github.com/jemtan/AI4LS_SummerSchool_UnsupervisedRepresentationLearning